

Addendum: J-Stream Tensor Autocorrelation as Third-Order Stylometric

Appended to: "A Universal Structural-Procedural Index: Emergent Identity, Meter and Rhythm in Music and Literature — Strange Loops in Bijective Braidings"

Third-Order Stylometric: Tensor Autocorrelation

The J-stream sign-flip rate (first tested as a candidate stylometric) proved uninformative — all works cluster near 0.5, the channel operating at entropy saturation. The tensor autocorrelation at lags 1-12 reveals what the scalar measure cannot: the periodic structure embedded in the ternary event sequence.

For each corpus, the autocorrelation of the full J stream and the nonzero-only J stream was computed at lags 1 through 12. The dominant period and its strength are reported.

Results:

Corpus	Dominant lag	Strength	Character
Bach Sarabande	7	0.1202	Strong periodic
Bach Prélude	11	0.1047	Strong periodic
Clair de Lune	3	0.0696	Moderate periodic
Under Milk Wood	1	0.0357	Near-noise
Jabberwocky	8	0.0325	Near-noise
Ode to Autumn	3	0.0249	Near-noise

The music/text boundary is unambiguous. Musical works produce periodic autocorrelation profiles with strength 0.07–0.12. Literary works produce near-noise profiles at 0.02–0.04. The tensor discriminates music from text at a level the J-stream distribution alone cannot.

The Sarabande. The slowest movement produces the strongest periodic signal — lag 7, strength 0.1202. Consistent with its triple metre and dotted rhythm. The Sarabande's

arithmetic stillness (D^* std deviation $3.26e-08$, near-zero) coexists with its strongest periodic structure. Stillness and periodicity are independent dimensions: the Sarabande is simultaneously the most spatially cohesive and the most temporally periodic movement in the corpus.

The Prélude. Dominant period at lag 11 (strength 0.1047) with a secondary signal at lag 3 (strength 0.1008). Bach's arpeggiated figures — 8 notes per bar, repeated — produce the lag 3 signal (the basic arpeggic unit). The lag 11 signal reflects cross-bar motivic structure. The mechanical repetition of the opening is arithmetically imprinted.

Clair de Lune. Dominant period at lag 3 (strength 0.0696). Clair de Lune is written in 9/8 — compound triple time, three groups of three. The tensor detected the triplet grouping without being told the time signature. Debussy's impressionist fluidity contains more periodic structure than expected — the 9/8 metre leaves its arithmetic trace.

Literary works. Keats' iambic pentameter — which might be expected to produce a lag 2 signal (the iambic foot) — registers only 0.0249. Carroll's rhyme scheme produces a weak lag 8 signal (0.0325). Dylan Thomas near-noise at lag 1 (0.0357). The tensor is detecting musical metre specifically. Linguistic metre, even as regular as iambic pentameter, does not produce comparable periodic structure in the J stream.

The predicted ordering confirmed:

Most periodic:	Bach Sarabande	(lag 7, strength 0.1202)
	Bach Prélude	(lag 11, strength 0.1047)
Musical middle:	Clair de Lune	(lag 3, strength 0.0696)
Text near-noise:	Thomas, Carroll, Keats	(0.02-0.04)

What the tensor adds to the stylometric portrait:

The existing portrait characterises works by clustering rate (structural repetition) and D^* std deviation (attractor tightness). The tensor autocorrelation adds a third independent dimension: temporal periodicity of the ternary event sequence. A work can score high on all three (Bach Suite — strong clustering, tight Sarabande, strong periodicity), or show independence across dimensions (Keats — low clustering, tight D^* , near-zero periodicity).

The three dimensions together constitute a more complete stylometric fingerprint than any single measure. No new machinery required — the J stream was already there.

This addendum supplements the Honest Scope section of the main note. The tensor autocorrelation was computed on the same J streams used for all other analyses. Full code available at unitambo.wordpress.com.

